

Swapnil Sanap

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PROFILE

Software Engineer with experience in Python, JavaScript, and Cloud Architecture. Proven ability to develop high-performance applications, ranging from Python-based face recognition systems to responsive web platforms. Proficient in utilizing AWS services (EC2, S3, RDS) for deployment and optimization. Adept at solving complex backend challenges while maintaining WCAG-compliant user interfaces.

EDUCATION

University of Illinois at Springfield (UIS) | Springfield, Illinois, USA

Master of Science in Computer Science

(Completed all required coursework toward M.S. in Computer Science)

Aug 2022 - May 2024

Prestige Institute of Engineering Management and Research (PIEMR) | Indore, India

Bachelor of Science in Computer Science (RGPV Affiliated)

July 2018 - Jun 2022

SKILLS

Languages & Front-End: Python, HTML5, CSS3, JavaScript (ES6+), React.js, React Native

Back-End: Node.js, Express.js, REST API integration

Databases: PostgreSQL, MySQL

Testing: Selenium, Jest

Tools & DevOps: Git, GitHub, CI/CD (basic), staging workflows, AWS

Performance & Optimization: SEO, Google Analytics

Machine Learning & Computer Vision: TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy

Other: JSON, Axios, responsive UI, version control

PROFESSIONAL EXPERIENCE

University Of Illinois at Springfield | Springfield, Illinois

Junior Web Specialist (Web Development)

July 2023 - May 2024

- Designed and implemented interactive front-end components using React.js, improving accessibility and mobile responsiveness across university sites.
- Developed custom modules and backend logic using Node.js and Express, streamlining admin tasks like content publishing and data sync.
- Contributed to the UIS Mobile App using React Native; optimized load times and resolved cross-platform layout issues.
- Integrated multiple REST APIs to dynamically fetch content, resulting in a 20% increase in data availability and performance.
- Collaborated with QA to implement Selenium and Jest test scripts, which cut down post-deployment bugs by 30%.
- Tuned Drupal's backend (minimal CMS work retained for context), optimized performance configs, and handled production rollouts through staging environments.
- Worked with Google Analytics and SEO tools to boost content visibility, tracking growth in engagement over a 3-month period.
- Ensured full WCAG 2.2 compliance, especially around forms, navigation, and contrast standards.

PROJECTS

SentinelVision - Real-Time Face Recognition System

April 2025 - May 2025

- Built a modular real-time face recognition pipeline using Python, OpenCV, and DeepFace for identity verification from live video feeds.
- Designed and implemented a preprocessing module to extract and cache face embeddings using Facenet, improving inference speed by ~4x.
- Integrated MTCNN for accurate face detection and cosine similarity for efficient facial matching against known embeddings.
- Enabled smooth live recognition through webcam/mobile IP stream with visual feedback and confidence scoring.
- Structured project into clear development phases (modular design), with plans for AI anomaly detection, alert logging, and a React-based dashboard.

Dmart Sales Prediction Software

Sep 2023 - Nov 2023

- Built a machine learning pipeline to predict Dmart retail sales trends based on historical data using Scikit-learn.
- Cleaned and transformed raw datasets with Pandas and NumPy to improve model training accuracy.
- Designed PostgreSQL schemas to store structured sales data for efficient querying and analysis.
- Tested and validated the model locally using sample datasets, documented insights and limitations.

Smart Menu Plus - Cloud-Enabled Restaurant App

Sep 2021 - Dec 2021

- Created a full-stack food ordering system with a responsive React.js UI for menu browsing and cart management.
- Built REST APIs using Node.js and Express to manage orders, inventory, and real-time cart updates.
- Implemented order tracking functionality for both users and restaurant staff, improving service flow.
- Focused on seamless UX with dynamic pricing, live order updates, and mobile responsiveness.

CERTIFICATE

- DevOps (PagerDuty)
- Apple Search Ads